

Phase 06 / v4 Report

Prompt-Level Conflict Detection in Qwen3-30B-A3B

DX Terminal / prompt_confusion / phase 06
2026-04-15

Dataset. `conflict_probe_examples_v4` (Neon), 768 rows, size-axis only. **Model.** Qwen/Qwen3-30B-A3B (vLLM, `resid_post` at last prompt token, 12 captured layers). **Run.** `wr_cc10418ff064_f8b538db`. Capture on Modal A100-80GB, analyses on Modal CPU, reporting local. End-to-end wall clock: ~10 min.

Abstract

We rebuilt the prompt-confusion dataset around a single conflict axis (trade-size directives), with four lexical variants each for STRATEGY and SETTINGS wordings and a 50/50 STRATEGY-first / SETTINGS-first order swap. Against this dataset, a linear probe on last-token residual activations decodes `conflict_present` at balanced accuracy ≥ 0.80 under every holdout condition tested, including a strict both-axes lexical holdout, while CountVectorizer + LogisticRegression on the raw prompt text is at chance. The probe signal shows a classic constructed-feature depth profile — chance at L0, rising through L16–L28, plateau from L28 onward.

A behavioral audit of the model’s generations surfaces a separate and important finding: the model’s **conflict resolution** (which side it follows) is dominated by surface wording format, not by a deeper arbitration mechanism. Settings written as a numeric scale (“Trade size: N/5...”) carry disproportionate authority compared to verbal-only setting wordings (“execution size can use the large tier”), and conditional strategy preambles interact with hedged market language to drive ~20% of aligned rows into refusal.

The detection result is robust and non-trivial. The resolution story is interesting as a descriptive artifact of v4 and motivates Phase 07 iteration, not a mechanistic claim in its own right.

Dataset

`conflict_probe_examples_v4` was a rebuild of v3 with four goals: restrict to one conflict axis (size) to eliminate the family-vocabulary confound of Phase 05; provide 4 lexical variants per axis (strategy and settings), up from 2 in v3; include a 50/50 STRATEGY-first / SETTINGS-first order split; and label refusal as a first-class outcome rather than dropping it.

Structural summary:

- **768 rows** — 384 conflict, 384 aligned matched pairs.
- **Families (2):** `trade_size_force_large`, `trade_size_force_small`. On conflict rows, SETTINGS prescribes the opposite size from STRATEGY.
- **Strategy variants (4 per direction):** e.g., `size_small_v0..v3`.
- **Setting phrase variants (4):** `size_setting_phrase_v0..v3`. Setting value 1 renders as “small” wording, value 5 as “large” wording.
- **Market pressure buckets (3):** `balanced`, `strategy_favored`, `setting_favored`, 2 context variants per bucket.
- **Section order:** 50/50 `strategy_first` / `setting_first`.

Lexical holdouts: variants v0/v1 are train, v2/v3 are test. Combined strict holdout tests both axes simultaneously.

Pre-capture QA ran text baselines on the generator output. All three text-baseline conditions hit `bal_acc = 0.50`, confirming no surface lexical leakage. Matched-pair integrity (384 pairs of 2) and per-cell class support (≥ 30 rows) passed.

Detection Results

Feature: `resid_post` at last prompt token (float16) across 12 layers. Probe: `SGDClassifier(loss="log_loss")` (`pipelines_v2.ProbeSpec`). Target: `conflict_present` (boolean). Baseline: `CountVectorizer(ngram 1-2) + LogisticRegression` on `user_text`.

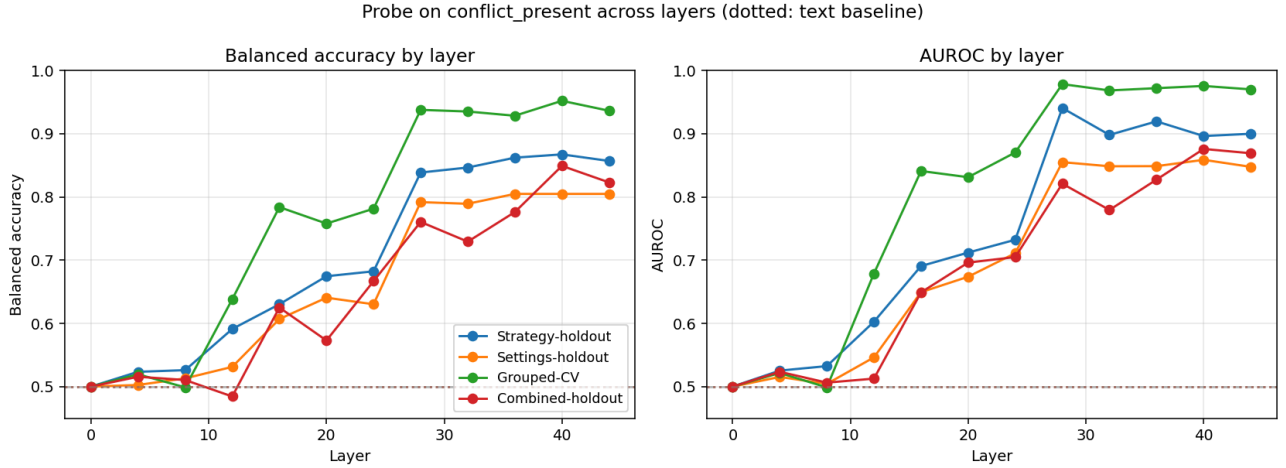


Figure 1: Probe performance on `conflict_present` across layers for four holdout conditions. Dotted lines mark the corresponding text baseline. Dashed gray line is chance.

Condition	Text bal / AUROC	Probe bal / AUROC	Peak layer
Grouped-CV (no holdout)	0.43 / 0.39	0.952 / 0.978	L40
Strategy-holdout (<code>v0/v1</code> → <code>v2/v3</code>)	0.50 / 0.50	0.867 / 0.941	L28–L40
Settings-holdout (<code>v0/v1</code> → <code>v2/v3</code>)	0.50 / 0.50	0.805 / 0.859	L36–L40
Combined strict holdout	0.50 / 0.50	0.849 / 0.876	L40

What this supports

The model builds a **linearly-decodable representation of STRATEGY / SETTINGS directive disagreement** at the last prompt token. The representation is **non-lexical** (text baselines at chance across all conditions, including the strict combined holdout where both strategy and setting wordings are unseen). It is a **constructed feature**: no signal at L0, steady ramp through L16–L28, plateau from L28. The model is computing the comparison, not pattern-matching.

Single-axis holdouts (0.81–0.87 peak) drop only modestly to the strict combined holdout (0.85). Both-axis leakage is a small effect, not a large inflation.

What this does not support

Nothing about how the model arbitrates conflicts — the probe target is `conflict_present`, not resolution direction. No causal claim that the detection representation drives downstream behavior. And no spatial localization: capture is at last prompt token, so the signal could be “detection,” “output-prep” content correlated with detection, or any linear combination.

Behavioral Audit

All 768 generations parsed cleanly as JSON. Labels derived by comparing output `size` against each row’s `strategy_expected_size` and `setting_expected_size`.

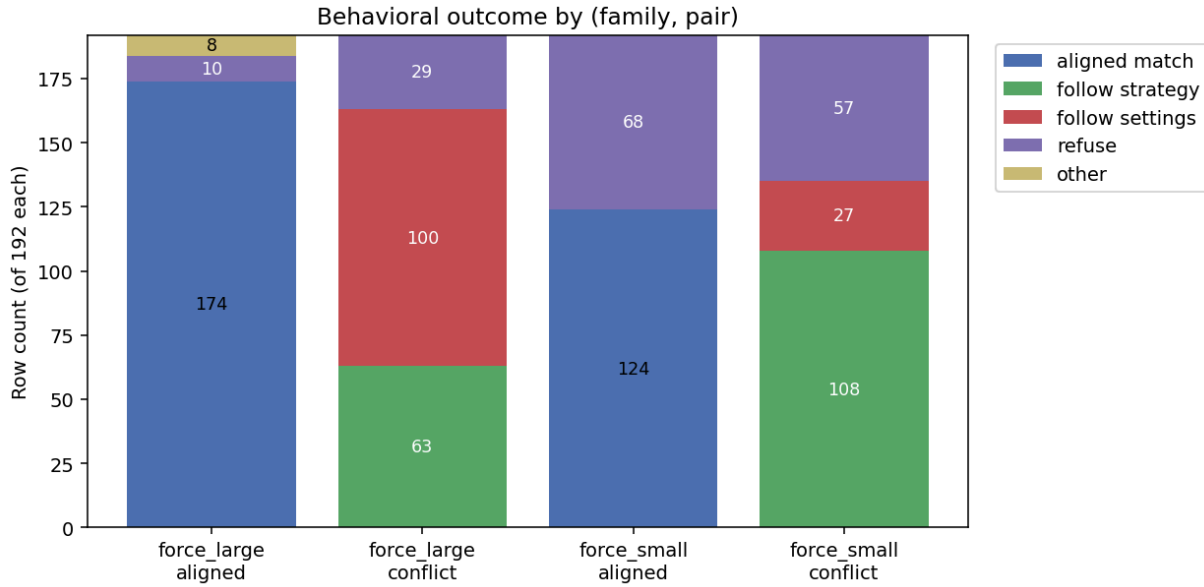


Figure 2: Behavioral outcome by (family, pair). Each column is 192 rows.

Overall distribution: `aligned_match` 298 (39%), `follow_strategy` 171 (22%), `follow_setting` 127 (17%), `refuse` 164 (21%), `other` 8 (1%).

Refusal is structured, not random

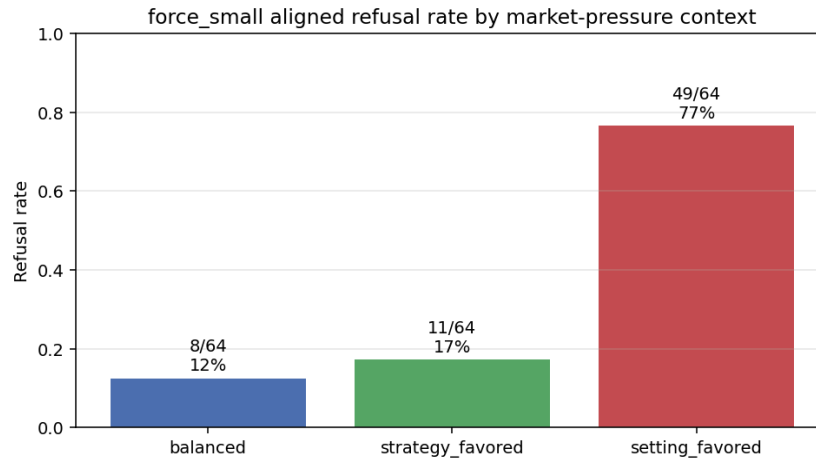


Figure 3: `force_small` aligned refusal rate by market-pressure context.

35% of `force_small` aligned rows refuse. These rows have `STRATEGY` and `SETTINGS` agreeing (“both say small”), so the refusal is not about conflict. It concentrates on the `setting_favored` market-pressure bucket, which has language like “usable edge with some confirmation, but one caution remains.” The combination of (a) conditional strategy wording (“*when an entry is justified*, use the small size tier”) and (b) explicit caution language in the market section is interpreted as “not justified” by the model. This is real multi-source reasoning, just on an axis different from the one we designed.

22% of conflict rows also refuse (15% `force_large`, 30% `force_small`). The system prompt explicitly allows refusal, so this is partly an instructed escape hatch.

Resolution is driven by wording format, not arbitration

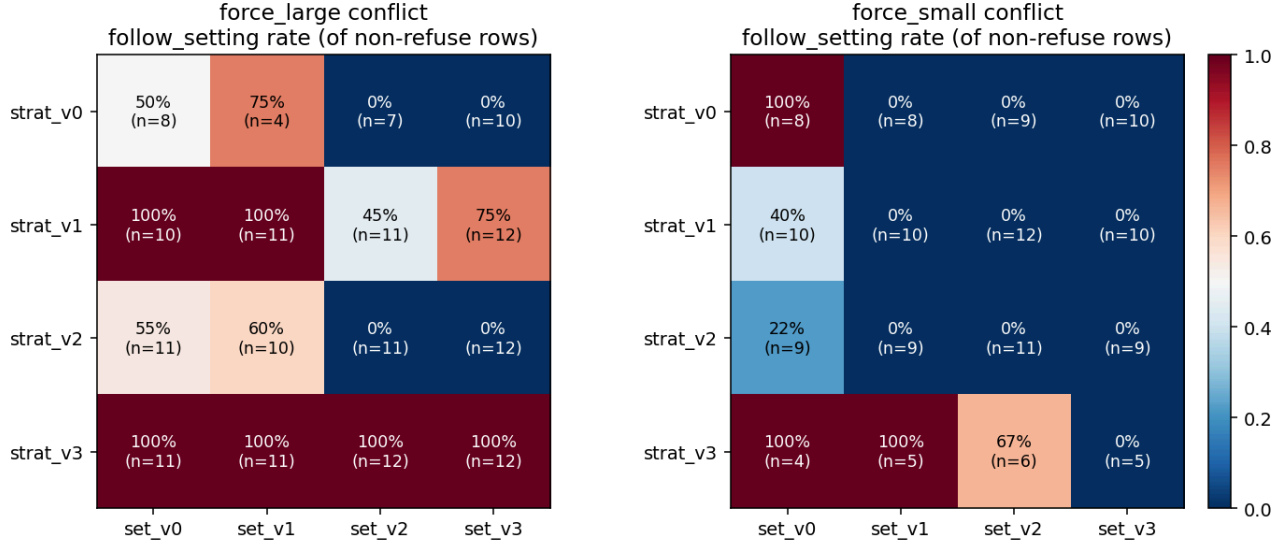


Figure 4: follow_setting rate by (strategy variant $\times$ setting variant) for conflict rows, computed over non-refuse outputs. Blue cells: model follows STRATEGY. Red cells: model follows SETTINGS.

The family-aggregate numbers (force_large: 52% follow_setting / 33% follow_strategy; force_small: 14% / 56%) average over variant combinations with qualitatively different behaviors.

Finding 1 (numeric-scale authority). On force_small conflict, setting_v0 (“Trade size: N/5. Use the [small|large] allocation tier.”) drives follow_setting in 22/24 non-refuse cases. The other three setting variants (verbal-only) drive follow_strategy in 86/92 non-refuse cases.

Finding 2 (variant v3 is soft). “On a clear setup, size up/down rather than scale down/up” is the softest strategy wording. On force_large conflict, strategy_v3 yields 0/46 follow_strategy across all four setting variants — the wording is effectively ignored.

Finding 3 (system prompt’s “SETTINGS still constrain” does not consistently win). Despite explicit instruction to prefer SETTINGS, STRATEGY wins in most verbal-setting cells. Resolution depends on whether SETTINGS is formatted as numeric authority or soft verbal guidance, not on the sys prompt directive.

What the Evidence Supports

Confident claims

The model builds a linearly-decodable representation of whether STRATEGY and SETTINGS contain contradictory sizing directives. This representation is non-lexical (survives combined holdout where both strategy and setting wordings are unseen), emerges in middle layers (L16–28 ramp), and stabilizes in upper layers (L28+ plateau). It is a **constructed feature**: the model is doing semantic comparison across two prompt sections, not pattern-matching on surface tokens. This is a real and non-trivial finding about prompt comprehension in Qwen3-30B-A3B.

Defensible, requires careful framing

The model’s conflict **resolution** behavior is dominated by wording format — numeric-scale settings carry disproportionate authority over verbal settings, and permissive/hedging language gets treated as optional. The defensible claim is “on this dataset, resolution is driven by surface formatting cues rather than

source identity. What we cannot yet claim is “**the model lacks a deeper arbitration mechanism**” — we have not tested a dataset where surface cues are controlled for.

Interesting, preliminary

The 77% aligned refusal on `force_small × setting_favored` shows genuine cross-section composition: the model is combining “*when an entry is justified*” with “*one caution remains*” and concluding no trade is justified. Multi-source reasoning is happening, just on an axis different from the one we designed for.

Not yet claimable

Nothing about how the model resolves conflicts internally — whether there is a representational signature that distinguishes “follow strategy” from “follow setting” outcomes. The probe target was `conflict_present`, not resolution direction. Nor can we say whether the detection representation **causes** downstream behavior or is a byproduct of comprehension the model routes through separately.

Novelty

The thread opened here — showing that format-mismatched sources produce shallow (surface-cue-driven) resolution while format-matched sources produce richer arbitration, and finding the mechanistic signature of that difference — is a genuinely novel claim if it holds. “Interpretability of policy conflict resolution as a function of prompt formatting” is directly relevant to real agent deployments where system prompts, tool outputs, and user instructions compete for behavioral authority in different formats.

Known Design Issues and Phase 07 Iteration

In rough priority for v5 iteration:

1. **Setting wordings duplicate scale semantics.** “Trade size: 1/5.
Use the small allocation tier.” embeds the scale gloss in every row. Scale should live in the system prompt once; SETTINGS lean (“Trade size: 1/5”). The disproportionate authority of `setting_v0` is partly driven by this in-line reinforcement.
2. **Conditional strategy preambles.** “When an entry is justified,” “if one asset clearly leads” give the model a conditional to evaluate against market context. On hedged markets this flips aligned rows into refusals. Decide deliberately whether conditionals are in scope (studying “does the entry fire at all?” is a different question from conflict resolution).
3. **System prompt pre-resolves the conflict.** “SETTINGS still
constrain the final execution” pre-answers the question we want to study. Remove or neutralize if resolution is a target claim.
4. **Refusal escape hatch.** “If no trade should be made, return
observe...” interacts badly with conditional strategies on hedged markets. Either keep and make hedged-market contexts unambiguously tradeable on aligned rows, or remove for a cleaner outcome space.
5. **Variant v3 is systematically softer than v0–v2.** Variant authoring for v5 should include a pre-publication behavioral check that all four variants produce similar aligned-match rates.
6. **Aggregated follow_strategy / follow_setting numbers are misleading.** Any resolution discussion must be broken out by (strategy variant × setting variant), because the variant-pair effect is the primary signal.

Operational Notes

End-to-end wall clock was ~10 min: 8m46s capture, 4 text baselines plus 4 probes in parallel on Modal CPU (~15–60s each), local report 17s. Capture config: `enforce_eager=False`, `max_num_seqs=16`, `enable_prefix_caching=True`, `enable_thinking=False`, `add_generation_prompt=True`, residual only.

Two repo fixes landed during this phase, documented in `docs/PIPELINES_V2_API.md` and the `constructing-workflows` skill: lazy-import `pipelines_v2.reporting` on the Modal import path so capture containers do not need `matplotlib`; and an `enable_thinking` kwarg on `VLLMEngine` to control the Qwen3 reasoning-mode chat template.

Reuse hazard observed: adding a derived label column to the dataset SQL changed the dataset semantic hash and forced `--reuse-completed` to re-run capture. Derived labels should land pre-capture or via `TransformSpec`, not via SQL column additions mid-phase.

Artifacts

- **Dataset** **generator:** `projects/DX_TERMINAL/prompt_confusion/phase_06/scripts/build_phase_06_dataset.py`
- **Workflow:** `specs/workflow.py` + `workflow.json` (same dir)
- **Auto-generated report (per-step charts):** `outputs/report/report_cc46450ff515_c31e2c79/`
- **Modal run:** `wr_cc10418ff064_f8b538db`. Capture artifact under `/artifacts/prompt_confusion/phase_06/capture_1_d5683901/` on the `xenon-data` volume.
- **Running notes:** `phase_06/notes.md`
- **Review snapshot:** `reports/review_2026-04-15.md`